MT Classes, Motihari

Tissues in Action

Class 9 Science, Chapter 3 (NCERT Exploration, new syllabus)



Meristems, permanent tissues, the four animal tissues, muscles, joints and the skeleton, explained with 31 textbook diagrams, five hands-on tools, every NCERT question solved and a quiz.

Start What is a tissue?

Every plant and animal starts as **one cell**. That cell divides again and again, and the new cells slowly become skin, muscle, bone, nerve, wood, leaf and everything else. This chapter is about how cells team up to do these different jobs.

Tissue

A **group of cells**, usually similar in structure, that **work together to perform a specific function**. Example: muscle tissue for movement, xylem for carrying water.

Levels of organisation

In a multicellular organism, the body is built in levels, like bricks → walls → rooms → building:

Cell (basic unit of life)

↓

Tissue (similar cells, one job)

↓

Organ (two or more tissues, e.g. heart, leaf)

↓

Organ system (organs working together, e.g. digestive system)



Why make tissues at all? Division of labour

In a unicellular organism like **Amoeba**, one cell does everything: eating, moving, breathing, removing waste. In a multicellular body, different groups of cells **specialise** in one job each. This sharing of work is called **division of labour**. A specialist does a job better than a "do-everything" cell, so the whole body works more efficiently and can carry out complex life processes.

Questions from the chapter opening

Try to answer first, then tap to check.

How is the study of cells and tissues significant for understanding life processes and human welfare?

Every life process (growth, movement, digestion, transport) is carried out by tissues. Knowing how they work helps doctors treat injuries and diseases (bone marrow transplant, healing fractures), helps farmers grow many healthy plants quickly by **tissue culture**, and helps scientists improve crops using tools like *Agrobacterium*.

How are tissues in plants and animals different, and why?

Plants stay in one place, make their own food and keep growing all their life, so they have rigid cell walls, supporting dead tissues and **meristems** for growth at fixed places. Animals move around in search of food, so they have flexible cells, muscles, nerves and connective tissue, and their growth is uniform rather than at fixed points. See Section 3.1.

How is division of labour at various levels of organisation correlated with structure and function?

At every level, the **shape fits the job**. Thin flat epithelial cells allow quick diffusion in the lungs; long, hollow, dead xylem cells form pipes for water; long muscle fibres can contract; long branched neurons carry messages far. Each tissue, organ and system does one part of the work, and together they keep the organism alive.

3.1 Why are plant and animal tissues different?

Plant and animal tissues are different because the two live very different lives. Compare them point by point:

Point	Plants	Animals
Movement	Fixed in one place; need to stand firm and upright.	Most move from place to place (sponges are an exception).
Cell wall	Rigid cell wall gives strength; many supporting tissues are **dead**.	No cell wall; cells can change shape, which helps in movement. Most tissues are **living**.
Nutrition	Tissues use sunlight to make food by **photosynthesis**.	Tissues **digest** food taken from outside.
Transport	Xylem (water) and phloem (food).	Blood carries food, gases and hormones.
Growth	Only in certain regions (meristems), and continues throughout life.	Growth happens all over the body and stops after a certain age.

Meet a scientist: B. G. L. Swamy

An Indian botanist known for his work on plant **morphology** (form) and **anatomy** (internal structure). His Kannada book ***Hasuru Honnu***, about plant life in the forests of the Western Ghats, won the **Kendra Sahitya Akademi Award in 1978**.

3.2 Meristems: tissues for growth in plants

A plant grows in three ways:

- it gets **longer** (stem grows taller, roots go deeper),
- it gets **thicker** (the stem's **girth** increases), and
- it **grows back** after cutting or grazing.

All three need cells that keep dividing. A tissue made of such actively dividing cells is called a **meristematic tissue** (meristem). There are three kinds, named after where they are found.

3.2.1 Apical meristem: how do plants grow in length?

Activity 3.1: The onion root experiment



Fig. 3.1 Onions on Jar A and Jar B

1. Place one onion bulb on each of two jars filled with water.
2. Measure the root length on days 1, 2 and 3.
3. On day 3, cut about 1 cm off the root tips in **Jar B**. Leave Jar A alone.
4. Keep measuring both for four more days.

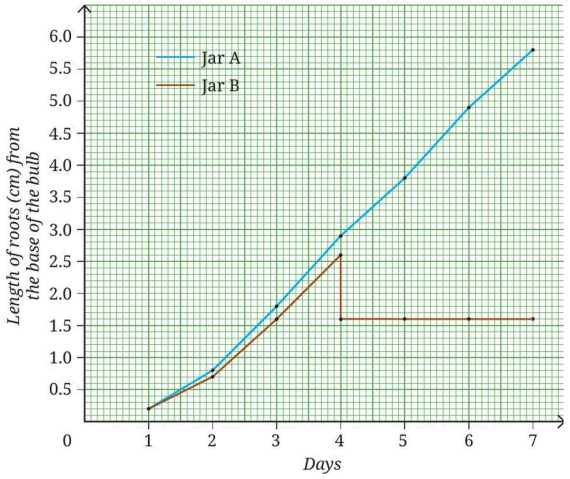


Fig. 3.2 Growth of onion roots. After the tips are cut, Jar B's line becomes flat.

Conclusion

Roots in Jar A keep growing; roots in Jar B **stop growing** once their tips are removed. So roots grow in length **only from their tips**, where cells divide continuously (you saw mitosis in onion root tips in Chapter 2). Shoot tips behave the same way. These growing zones at the tips of roots and shoots are the **apical meristems** (apex = tip).

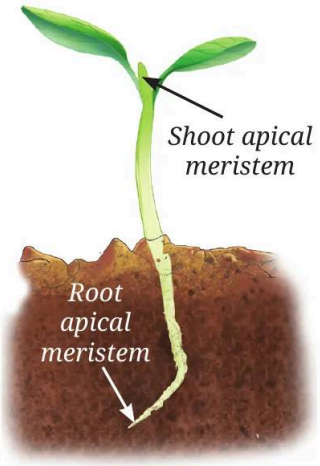


Fig. 3.3 Apical meristem at the shoot tip and the root tip.

3.2.2 Lateral meristem: how do plants grow in girth?

Stems of dicot plants (like mango, neem, teak) do not just grow taller; they also get **thicker** every year. In the stem there is a **ring** of dividing cells. They add new cells both towards the inside and towards the outside, in circles, so the stem's diameter grows. This side (lateral) growing tissue is the **lateral meristem**.



Fig. 3.4 T.S. of a tree trunk with annual rings

🌳 Annual rings

The lateral meristem makes one ring of wood each year. **Count the rings to find the age** of the tree. A **wide** ring means a good year (enough rain, warmth); a **narrow** ring means a hard year (drought, cold). So rings also record the climate the tree lived through.

3.2.3 Intercalary meristem: how do plants grow back after cutting?

If you cut the tip of a young stem, it stops getting taller, but **new branches come out from the nodes**. A **node** is the point on a stem where leaves or branches arise; the part of the stem between two nodes is an **internode**. The **intercalary meristem** lies at the **base of the internode** (just above a node). That is why a trimmed hedge becomes bushy and a mowed lawn or grazed grass grows back again.



Fig. 3.5 New branches from the nodes of a cut stem



Fig. 3.6 Lawn mowing: grass grows back from intercalary meristems

Meristem	Where?	What does it do?
Apical	Tips of roots and shoots	Increases **length**
Lateral	In a ring along the circumference of stems (and roots)	Increases **girth** (thickness)
Intercalary	Base of internodes / at nodes (e.g. grasses)	**Regrowth** after cutting or grazing; branching

Features of meristematic cells

Small cells with thin walls easy to divide

A thin wall does not get in the way of splitting into two.

Large, prominent nucleus controls division

The nucleus holds the DNA that must be copied every time the cell divides.

Dense cytoplasm, many organelles busy factory

Lots of material to build new cells quickly.

Vacuoles generally absent no storage

The cell is not storing anything yet, and its space is needed for active cytoplasm.

Tightly packed, no intercellular spaces

New cells are packed side by side as they form.

📝 Why do meristem cells lack vacuoles? (asked in the text)

Vacuoles are mainly for **storage** (water, food, waste). Meristem cells are not storing; they spend all their energy on **dividing**. A big vacuole would fill the cell and push the dense cytoplasm aside, slowing division. Vacuoles form later, when the cell stops dividing and specialises.

From meristem to permanent tissue: differentiation

Some new cells made by a meristem stay meristematic. Others **lose the ability to divide**, change their shape and structure and take up one job (support, transport or storage). These cells form **permanent tissues**.

Differentiation

The process by which cells of meristematic tissue become specialised in structure and function, forming permanent tissue.

Meristematic tissue → differentiation → Permanent tissue

3.2.4 Permanent tissues

If you look at a thin transverse section (T.S., a cross-section) of a sunflower stem under a microscope, you see many different kinds of cells in neat layers. Each layer is a permanent tissue with its own job.

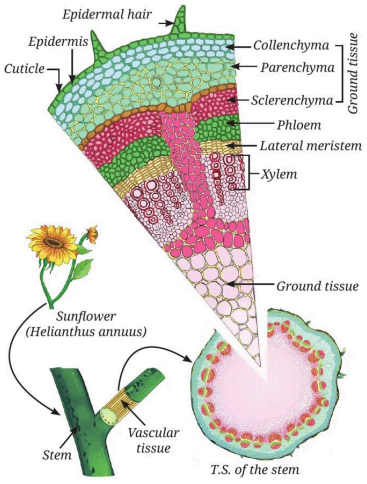


Fig. 3.7 Internal structure of a sunflower stem. From outside to inside: cuticle and epidermis → collenchyma → parenchyma → sclerenchyma → phloem → lateral meristem → xylem → ground tissue (pith) in the centre.

Two kinds of permanent tissue

Simple permanent tissue: made of **only one type** of cell (parenchyma, collenchyma, sclerenchyma).
Complex permanent tissue: made of **more than one type** of cell working together (xylem, phloem).

(i) Protective tissue: epidermis

The **epidermis** is the outermost layer of all plant parts. It is a **single layer** of flat, rectangular, tightly packed cells with no gaps, so germs and water cannot easily pass.

- **Cuticle:** a waxy layer of **cutin** on the epidermis. It cuts down water loss and protects from injury and parasites. Desert plants have a **thick** cuticle.
- **Hairs:** in roots, epidermal cells grow into **root hairs** that increase the surface area for absorbing water and minerals.
- **Stomata:** tiny pores in the leaf epidermis for **gas exchange** and **transpiration** (loss of water vapour). Transpiration creates a **pull** that draws water up the xylem, and also helps remove some wastes.

🌲 **Ready to go beyond: cork (bark)**

Young plants have a one-cell-thick epidermis. In older stems, some cells under the epidermis start dividing again and become the **cork cambium** (a lateral meristem). It produces **cork cells**: dead, tightly packed, with a substance that makes them waterproof and airtight. Cork forms the **bark** of a tree.

(ii) Supporting tissue: simple permanent tissues

Why does a plant stay upright? Why does a fresh twig bend but a dry twig snap? Why are seed coats hard, and how do water plants float? Three simple tissues answer these:

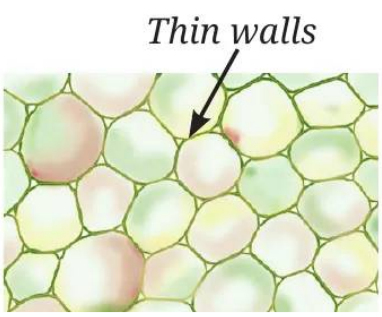


Fig. 3.8 (a) Parenchyma

Living, **thin walls**, loosely packed with spaces. **Stores food**; does photosynthesis in green parts; air-filled parenchyma helps water plants **float**.

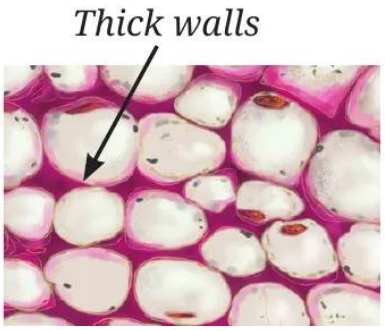


Fig. 3.8 (b) Collenchyma

Living, walls **thickened at the corners** by **pectin** (rubber-like). Gives **support with flexibility**, so stems and tendrils bend without breaking.

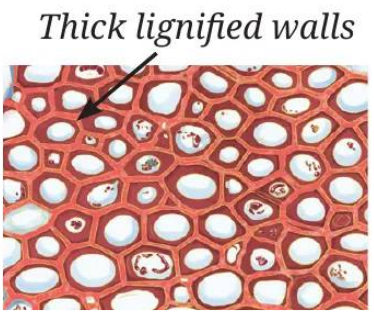


Fig. 3.8 (c) Sclerenchyma

Mostly **dead**, very **thick walls** of **lignin**. Hard and strong (woody). Found in stems, leaf veins, seed coats, **coconut husk** and walnut shell.

Feature	Parenchyma	Collenchyma	Sclerenchyma
Living or dead	Living	Living	Mostly dead
Cell wall	Thin	Thick at corners (pectin)	Thick all round (lignin)
Intercellular space	Present	Very little	Absent
Main job	Storage, photosynthesis, floating	Flexible support	Strength, hardness
Example	Potato, pith, leaf	Leaf stalk, young stem, tendrils	Coconut husk, jute fibre, nut shells

📝 **Memory trick**

Parenchyma = **P**antry (stores food). **C**ollenchyma = **C**orners (and flexible). **S**clerenchyma = **S**trong and **S**tiff (and dead).

(iii) Conducting tissues: complex permanent tissues

How does water reach the top of a tall tree, and how does sugar made in a leaf reach the roots? Through two pipe systems, **xylem** and **phloem**. They are called **complex** tissues because each is made of **several different cell types** working together.

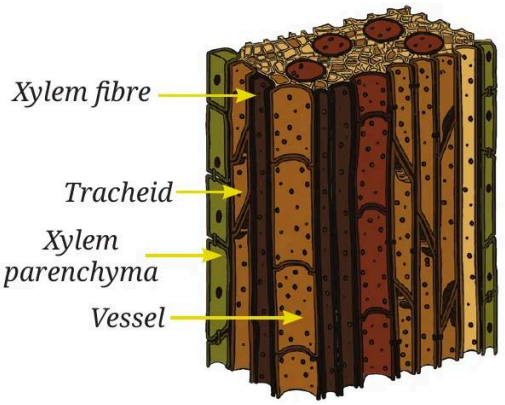


Fig. 3.9 (a) Xylem: carries water and minerals upward

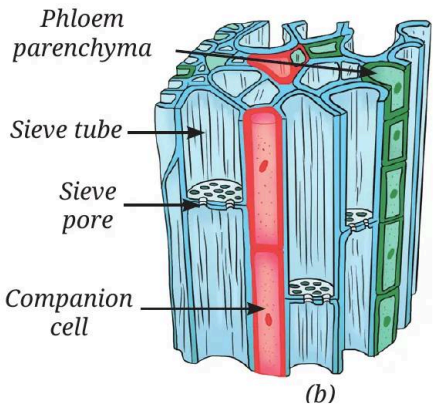


Fig. 3.9 (b) Phloem: carries food in all directions

	Xylem	Phloem
Carries	Water and minerals, **from roots up** to other parts	Food (sugars), **from leaves** to other parts (up and down)
Parts	Tracheids, vessels, xylem parenchyma, xylem fibres	Sieve tubes, companion cells, phloem parenchyma, phloem fibres
Living or dead	Mostly **dead**. Only **xylem parenchyma** is living.	Mostly **living**. Only **phloem fibres** are dead.
Pipes	Tracheids and vessels: tubular, thick-walled	Sieve tubes: long cells joined end to end through perforated walls (sieve plates)
Extra job	Gives the plant strength	Phloem parenchyma stores food, resin, tannins, latex

Who does what in phloem?

- **Sieve tubes**: the actual pipes that carry food.
- **Companion cells**: special parenchyma cells beside the sieve tubes. They control the sieve tube cells and manage the **loading and unloading of sugars**.
- **Phloem parenchyma**: storage. **Phloem fibres**: dead, sclerenchyma-like, give strength.

📝 Don't mix them up

Xylem carries water, phloem carries **f**ood: "**Ph**loem = **F**ood" (both sound like "f"). Xylem moves **one way** (up); phloem moves **both ways**.

Systems

Plant tissue systems

Tissues in a plant do not work alone. They are grouped into three larger **tissue systems**, found in the root, stem and leaf alike:

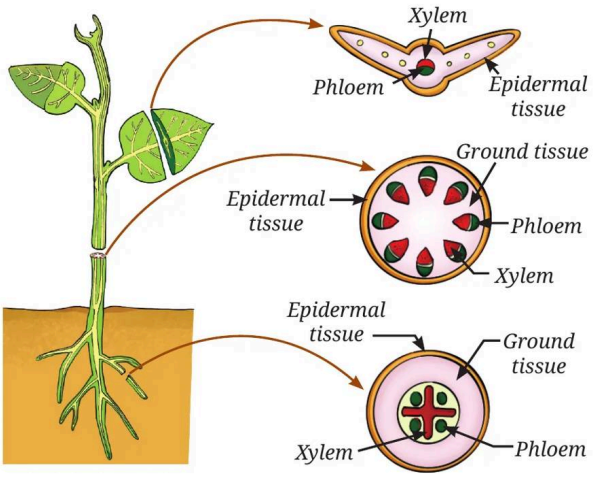


Fig. 3.10 Tissue systems in the leaf (top), stem (middle) and root (bottom). Note how xylem and phloem are arranged differently in each.

1. Dermal tissue system the skin

Outer covering (epidermis, later cork). Protects inner parts and reduces water loss.

2. Ground tissue system the filling

The main body between the skin and the pipes. Made of parenchyma, collenchyma and sclerenchyma.

3. Vascular tissue system the plumbing

The conducting tissues, xylem and phloem.

Plant tissues at a glance

Meristematic (can divide)	Permanent (cannot divide)
Apical tips: length	**Protective** epidermis (cuticle, hairs, stomata); cork
Lateral ring in stem: girth	**Simple / supporting** parenchyma, collenchyma, sclerenchyma
Intercalary	**Complex / conducting**

nodes/base of internode: regrowth

xylem, phloem

Meet the scientists: Sipra Guha Mukherjee and S. C. Maheshwari

These Indian scientists made a breakthrough in **plant tissue culture**: they grew complete plants from **anther culture** (pollen-bearing part of a flower) on an artificial nutrient medium in the lab. Their work helped crop improvement and modern agriculture.

3.3 Animal tissues: epithelial tissue

Blink. Make a fist. Take a deep breath. Touch something cold. Each of these uses a different animal tissue. Animals have **four main types** of tissue:

E	C	M	N
Epithelial *covers and lines*	Connective *connects and supports*	Muscular *moves*	Nervous *controls*

3.3.1 Epithelial tissue: structure and functions

Epithelial tissue forms the outer covering of the body (skin) and lines the inside of organs such as the mouth, lungs, blood vessels and intestine. Its cells are **closely packed** with almost no space between them. This keeps germs out, reduces water loss, and controls what is absorbed, secreted or passed across.

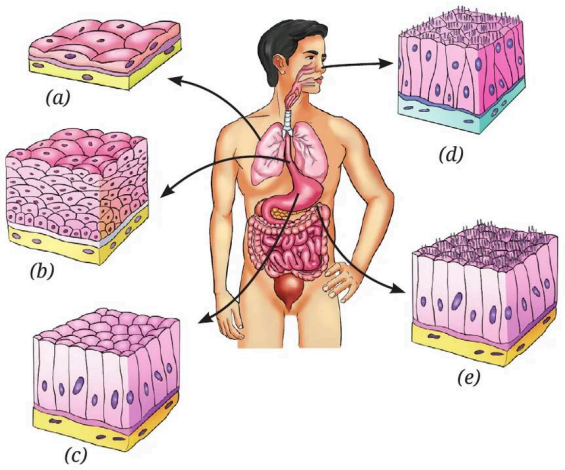


Fig. 3.11 Types of epithelial tissue: (a) thin flat cells, (b) many layers, (c) cuboidal/columnar secreting cells, (d) sensory cells with cilia, (e) tall pillar-like absorbing cells.

Function	Structure	Where

Exchange: fast diffusion of gases and liquids	One layer of thin, flat cells (a)	Blood vessels, lungs (air sacs)
Protection: from injury, friction, germs	Many layers; outer cells flat and tightly packed (b)	Skin, mouth, oesophagus
Secretion: mucus, enzymes, hormones, sweat, saliva	Cuboidal or columnar gland cells (c)	Salivary glands, sweat glands, stomach lining
Sensing: smell, taste, sound, balance	Receptor cells with hair-like cilia (d)	Nose, taste buds, inner ear
Absorption: nutrients and water	One layer of tall, pillar-like cells, often with tiny hair-like projections (e)	Lining of the small intestine

Structure fits function

Where things must **pass through quickly** (lungs, blood vessels), the epithelium is **thin, one layer**. Where the body must be **protected from rubbing** (skin, food pipe), it is **many layers thick**.

3.3.2 Connective tissue

A tissue that **connects and supports** other tissues is a **connective tissue**. Blood and bone are both connective tissues, yet blood is a liquid and bone is rock-hard. The difference lies in the **matrix**, the material in which the cells are embedded:

- **Blood**: matrix is a watery fluid (plasma).
- **Bone**: matrix is hard and rigid (calcium and phosphorus compounds).
- **Cartilage**: matrix is soft and jelly-like, so it bends.

Blood: a fluid connective tissue

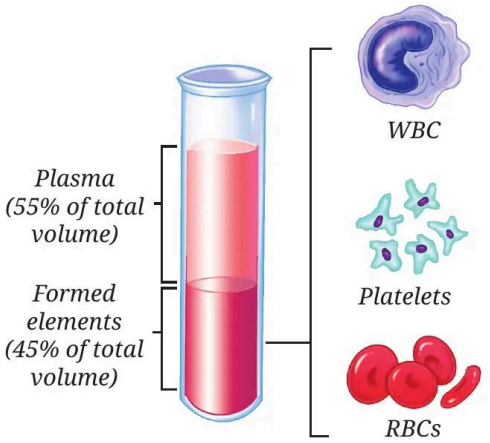


Fig. 3.12 (a) Components of blood: plasma (55%) and formed elements (45%).

Activity 3.2: Blood in everyday life

Experience	What you notice	Which part of blood explains it?
A small cut on the skin	Red blood oozes out; a clot forms after some time	Red colour: **haemoglobin** (iron-rich protein) in **RBCs**. Clot: **platelets**.
A skin infection	Area turns red and swollen; maybe fever and pus	**WBCs** gather at the infection to fight germs, causing inflammation and pus.
Exercise or running	Faster breathing; face turns red	Muscles need more **oxygen**; more blood flows, carried by **RBCs**.

Remember

- **RBCs** carry oxygen (haemoglobin); live about **4 months** (about 120 days) and are replaced regularly.
- **WBCs** fight infection.
- **Platelets** help blood clot.
- **Plasma** carries dissolved food, hormones, wastes.

Bone, cartilage, tendon and ligament

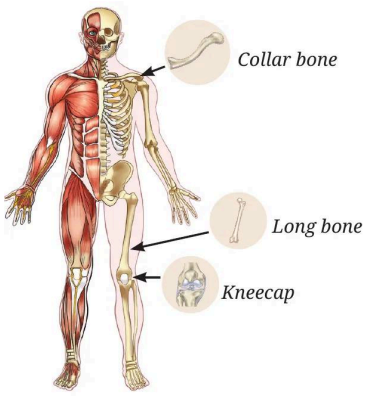


Fig. 3.12 (b) Types of bones: collar bone, long bone, kneecap

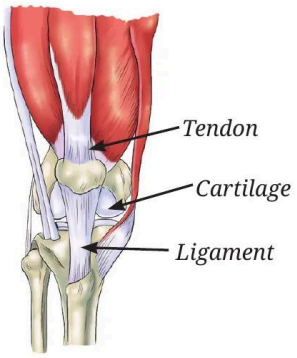


Fig. 3.12 (c) Arrangement of tissues at a joint (knee)

Activity 3.3: Feel your connective tissues

Do this	You feel	Job	Tissue
Touch your elbow	Hard, rigid	Strength, support, protection	**Bone**
Fold your ear / press your nose tip	Soft, bends, springs back	Flexibility; cushions bone ends, absorbs shock	**Cartilage**
Hold your forearm and wiggle fingers	Movement in the forearm	Joins **muscle to bone**; makes movement happen	**Tendon**
Raise your leg as far as the knee allows	Joint stops at a limit	Joins **bone to bone**; gives stability, prevents dislocation	**Ligament**

📝 **Tendon or ligament?**

Tendon: muscle **T**o bone (tendon = "muscle-tie"). **L**igament: bone to bone, **L**imits movement. This is a favourite exam trap (see Assertion-Reason C).

3.3.3 Muscular tissue: can we control movement?

Some movements we control (running, writing, lifting): these are **voluntary**. Others happen on their own (heartbeat, food moving in the intestine): these are **involuntary**. Three kinds of muscle do this work.

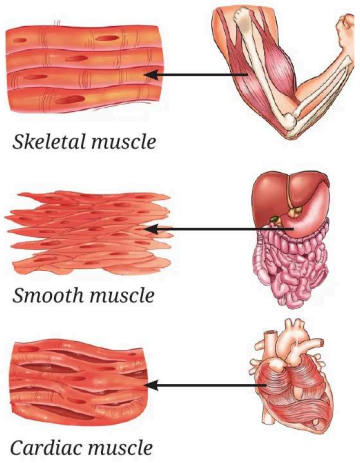


Fig. 3.13 Different types of muscles

	Skeletal	Smooth	Cardiac
Where	Attached to bones	Stomach, intestine	Only heart
Control	Voluntary	Involuntary	Involuntary
Shape	Long, cylindrical, **unbranched**	**Spindle**-shaped	Cylindrical, **branched**
Nuclei	**Many** (multinucleate)	One	One
Striations	Yes (light and dark bands)	No	Faint
Work	Quick movement, tires	Slow, continuous	Rhythmic, never tires

3.3.4 Nervous tissue: sensing and responding

Pulling your hand away from a hot pan, or remembering an old song: both are the work of **nervous tissue**, the body's control and communication network. The **brain** is the control centre. Muscles cannot work on their own; they wait for instructions from nerves. (During exercise, the brain tells the heart to beat faster.)

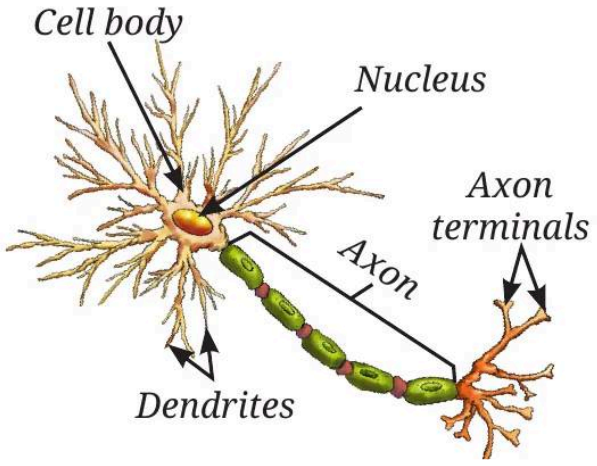


Fig. 3.14 Structure of a neuron

1. **Dendrites**: short branches that **receive** signals from other neurons.
2. **Cell body**: contains the **nucleus**; controls the cell.
3. **Axon**: a long fibre that **carries** the message away from the cell body.
4. **Axon terminals**: **pass** the message on to the next cell (neuron, muscle or gland).

📝 **Direction of a message**

Dendrite → cell body → axon → axon terminal. Remember "**D**o **C**ats **A**lways **T**alk?"

3.4 The musculoskeletal system

The **musculoskeletal system** = bones + muscles + joints + cartilage + tendons + ligaments. It lets us stand upright, move, keep our posture and protect delicate organs.

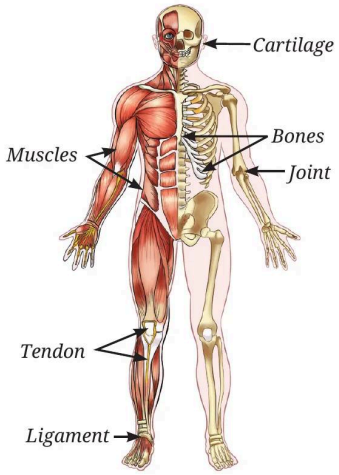


Fig. 3.15 Musculoskeletal system

Nerve sends a signal

↓

Muscle contracts (shortens)

↓

Tendon passes the pull to the bone

↓

Bone moves at a joint

Muscles can only **pull**, never push. The whole system works under the control of the nervous system.

3.5 Types of joints

A **joint** is a place where two or more bones meet. Joints allow movement, but cannot move bones by themselves: **muscles** pull the bones. How far a joint can move depends on the shape of the bone ends.

Activity 3.5: Which way does each part move?

Body part	Complete rotation	Partial rotation	Bending (up-down / side)	Joint
Elbow	No	No	Yes	Hinge
Shoulder	Yes	Yes	Yes	Ball and socket
Knee	No	No	Yes	Hinge

Neck	No	Yes	Yes	Pivot
Fingers	No	No	Yes	Hinge
Toes	No	No	Yes	Hinge
Wrist	No	Yes	Yes	Gliding (side to side, up-down)

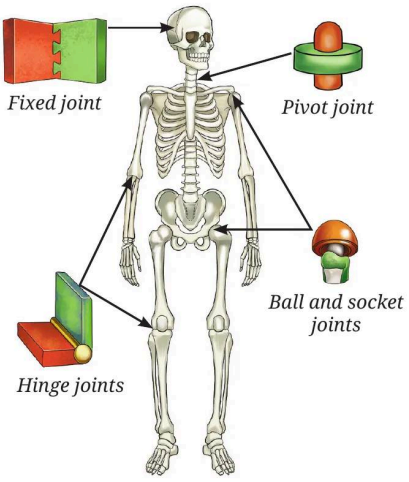


Fig. 3.16 Types of joints and where they are in the skeleton

3.5.1 Ball and socket joint shoulder, hip

The rounded head of the upper arm bone sits in a shallow cup of the shoulder bone. Movement: forward, backward, sideways and **all the way round**. With the collarbone, the shoulder forms the **shoulder girdle**, which joins the arm to the skeleton.

3.5.2 Hinge joint elbow, knee, fingers

Bends and straightens in **one direction only**, like a door hinge. The knee is protected by a small bone, the **kneecap**.

3.5.3 Pivot joint neck (skull on backbone)

One bone turns around another, like a doorknob. Lets you shake your head "no" (side to side).

3.5.4 Fixed joint skull

The flat skull bones are locked together and **cannot move**. This keeps the brain safe even when the body moves.



Fig. 3.17 Poses of classical and folk dances of India (see Pause and Ponder 5)

🩸 Ready to go beyond: stem cells

Bone marrow contains **stem cells** that can divide and make new blood cells. In a **bone marrow transplant**, stem cells from a healthy donor are given to patients with blood cancers such as **leukaemia** or disorders such as **thalassaemia**.

3.6 Skeletal system

The **skeletal system** is the body's framework of bones. It gives strength and protects soft organs. Its main parts:

Skull protects brain, eyes, ears

Flat bones joined by fixed joints.

Backbone (vertebral column, spine) supports, keeps us upright

A flexible chain of small bones called **vertebrae**, starting from the base of the skull. A **cartilage disc** between each pair works as a cushion and lets us bend and twist without hurting the **spinal cord** inside.

Rib cage protects heart and lungs

12 pairs of ribs, joined to the backbone at the back and the **breastbone (sternum)** in front by flexible **cartilage**. This flexibility lets the cage expand and shrink as we breathe, moving air in and out of the lungs. That is why a rib injury makes breathing painful.

🧘 Bridging science and society: Yoga

Yoga, from ancient Indian texts, combines postures, breathing and meditation. It improves flexibility, posture and breathing, lowers stress and helps prevent lifestyle diseases. **21 June** is **International Yoga Day**. Good posture, food, exercise and yoga keep bones strong, muscles fit and joints flexible.

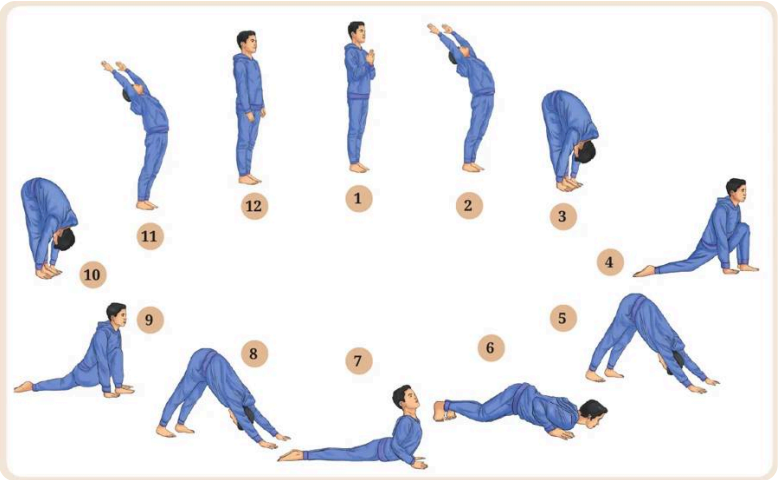


Fig. 3.18 The 12 poses of Surya Namaskar. Try naming the joints that bend in each pose.

Lab

Totipotency and tissue culture

Think as a scientist: one cell → a whole plant

In **1958**, **F. C. Steward** took cells from the **phloem of a carrot** and grew them in a nutrient liquid with sugars and hormones. The cells divided into a lump, and the lump slowly grew into a **complete carrot plant**. He was the first to do this.

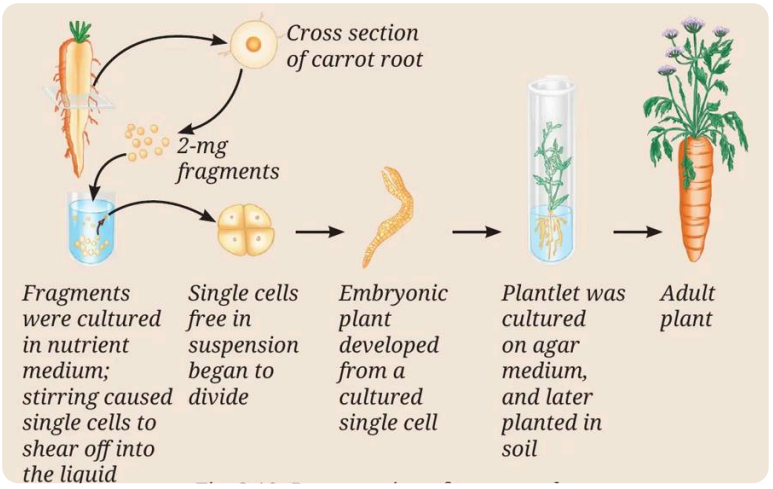


Fig. 3.19 Regeneration of a carrot plant from single phloem cells.

Mature phloem cell (specialised, not dividing)

↓

Dedifferentiation: it regains the ability to divide

↓

Mass of unspecialised cells

↓

Redifferentiation: forms roots, shoot, embryo

↓

Complete new plant

Totipotency

The ability of some mature plant cells to undifferentiate, divide and redifferentiate into a complete new plant under suitable conditions. Such cells are **totipotent**, just like a zygote, which divides and differentiates into a whole organism.

Table 3.6: Steward's results

Light	Air	Medium	Change in fresh weight
✓	✗	Solid + nutrients	Reduced
✓	✓	Liquid + nutrients	**20% increase**
✗	✓	Liquid + nutrients	Reduced

– **(a) What do you conclude about the phloem cells of carrot?**

They are **totipotent**: even though they are mature and specialised, they can go back to dividing (dedifferentiate) and then redifferentiate into all the parts of a complete plant.

– **(b) Which combination gives the highest and the lowest biomass, and why?**

Highest: light + air + liquid medium (20% increase). Light allows photosynthesis, air supplies oxygen for respiration (energy), and a liquid medium lets every cell touch the nutrients and stay separate while stirring.
Lowest: the other two both lost weight. In the solid medium without air, cells lack **oxygen** for respiration and get fewer nutrients; in the dark, cells cannot photosynthesise and use up their stored food. The table does not give exact numbers, but the no-air set-up is likely the worst because without oxygen cells cannot release energy at all.

(c) Would animal cells give the same result?

No. Most mature animal cells lose the ability to form a whole organism. They can divide in culture to make more cells of the same kind, but they do not redifferentiate into a complete animal. Only the zygote (and, to a limited extent, stem cells) can do that. Animal cloning needs special methods such as putting a body-cell nucleus into an egg cell.

(d) Two commercial applications?

1. **Micropropagation**: producing thousands of identical, disease-free plants quickly from a few cells (banana, orchids, sugarcane, potato).
2. Saving rare or endangered plants, and making **genetically improved crops** or medicinal chemicals from cultured plant cells.

Bridging science and society: crown gall



Fig. 3.20 Crown gall disease

Crown gall is a plant disease with tumour-like swellings on stems, caused by **uncontrolled cell division**. It is caused by the bacterium ***Agrobacterium tumefaciens***, which inserts its own genes into plant cells.

Scientists turned this trick into a tool: today *Agrobacterium* is used to put **useful genes** into plants to make better crops, disease-resistant varieties and valuable plant chemicals.

Recap At a glance

- A **tissue** is a group of similar cells working together for a specific function. Tissues bring **division of labour**.
- Plant tissues: **meristematic** (can divide) and **permanent** (cannot divide).
- Meristems: **apical** (length), **lateral** (girth, annual rings), **intercalary** (regrowth). Their cells are small, thin-walled, with big nuclei, dense cytoplasm and no vacuoles.
- **Differentiation** turns meristem cells into permanent tissue.

- By function: **protective** (epidermis, cork), **supporting** (parenchyma, collenchyma, sclerenchyma), **conducting** (xylem, phloem).
- **Simple** tissue = one cell type; **complex** tissue = several cell types (xylem, phloem).
- Xylem: water and minerals upward, mostly dead. Phloem: food both ways, mostly living.
- Three plant tissue systems: **dermal, ground, vascular**.
- Animal tissues: **epithelial** (covers and lines), **connective** (blood, bone, cartilage, tendon, ligament), **muscular** (skeletal, smooth, cardiac), **nervous** (neurons).
- Tendon: muscle to bone. Ligament: bone to bone.
- Joints: ball and socket, hinge, pivot, fixed.
- Skeleton = skull + backbone (vertebrae) + rib cage (12 pairs). Movement = muscles + bones, controlled by the nervous system.
- Some plant cells are **totipotent**: one cell can regrow a whole plant (tissue culture).

🔭 The journey beyond

- Ask a doctor about ligament tears, cartilage damage and fractures, and how food and lifestyle reduce the risk.
- Sit with feet flat, hold the back of your ankle just above the heel and point your toes up and down: you will feel the **Achilles tendon** move. Find tendons at other joints.
- Think how yoga or kabaddi builds bone and muscle; how pruning, grafting and irrigation help meristems and conducting tissues.
- On a nature walk, compare leaves from dry, wet and water habitats (thick cuticle? waxy? floating?). Ask elders about leaves used for plates, cooling wraps or keeping insects away.
- Learn five steps of a tribal dance and list the joints each step uses.



Fig. 3.24 Feeling the ankle tendon



Fig. 3.25 A tribal dance

✨ **The quest continues: can we get a complete animal from one animal cell, like plants? Advantages and challenges?**

Not by simply culturing a body cell, because mature animal cells are not totipotent. It has been done only by **cloning** (e.g. Dolly the sheep, 1996), where the nucleus of a body cell is put into an egg cell whose own nucleus is removed, and the embryo is grown in a mother.
Advantages: saving endangered animals, copying high-yield farm animals, growing tissues and organs for transplant from a patient's own cells.
Challenges: very low success rate, health problems in clones, high cost, loss of genetic variety, and ethical questions.

Quiz

Test yourself

Tap an option to see the right answer and the reason.

1. Grass grows back after grazing because of the

- apical meristem
- lateral meristem
- **intercalary meristem**
- epidermis

Intercalary meristem at the nodes / base of internodes lets grass regrow after its tips are eaten.

2. The increase in girth of a tree trunk is due to

- apical meristem
- **lateral meristem**
- intercalary meristem
- cork

A ring of lateral meristem adds new cells inside and outside, making annual rings.

3. Which tissue has living cells with walls thickened at the corners by pectin?

- Parenchyma
- **Collenchyma**
- Sclerenchyma
- Xylem

Collenchyma: corner thickening with pectin gives flexible support.

4. The only living component of xylem is

- tracheid
- vessel
- **xylem parenchyma**
- xylem fibre

Tracheids, vessels and fibres are dead; xylem parenchyma is living.

5. Loading and unloading of sugar in sieve tubes is managed by

- **companion cells**
- phloem fibres
- vessels
- guard cells

Companion cells are specialised parenchyma cells beside sieve tubes.

6. Coconut husk fibres are made of

- parenchyma
- collenchyma
- **sclerenchyma**
- phloem parenchyma

Dead cells with thick lignified walls make the husk hard and fibrous.

7. Epithelium of the small intestine that absorbs food is made of

- many layers of flat cells
- **one layer of tall pillar-like cells**
- cells with thick lignin walls
- branched cells

Tall columnar cells with hair-like projections give a large surface for absorption.

8. Bone is joined to bone by

- tendon
- **ligament**
- cartilage
- muscle

Ligaments join bones and limit movement; tendons join muscle to bone.

9. Muscle fibres that are branched, have one nucleus and faint striations are

- skeletal
- smooth
- **cardiac**
- voluntary

Cardiac muscle, found only in the heart, works rhythmically without tiring.

10. The part of a neuron that receives signals from other neurons is the

- axon
- **dendrite**
- axon terminal
- nucleus

Dendrites receive; the axon carries the message away; terminals pass it on.

11. Shaking the head to say "no" uses a

- hinge joint
- fixed joint
- **pivot joint**
- ball and socket joint

The skull turns on the backbone at a pivot joint, like a doorknob.

12. Plasma makes up about what share of blood volume?

- 25%
- 45%
- **55%**
- 90%

Plasma about 55%; formed elements (RBCs, WBCs, platelets) about 45%.

13. The ability of a single carrot phloem cell to form a whole plant is called

- differentiation
- **totipotency**
- transpiration
- division of labour

Totipotent cells can dedifferentiate, divide and redifferentiate into a full plant (F. C. Steward, 1958).

14. The rib cage can expand during breathing because ribs are joined to the sternum by

- tendons
- fixed joints
- **flexible cartilage**
- smooth muscle

Cartilage is flexible, so the cage can expand and contract.

NCERT

Answers to all questions

Tap a question to see the answer. Use **Print or save as PDF** at the top to get all answers open.

Pause and Ponder

1. Coconut husk fibres are hard and brittle, but coriander leaf stalks are soft and flexible. Why?

Coconut husk is made of **sclerenchyma**: dead cells with thick walls of **lignin**, so it is hard and stiff. A coriander leaf stalk has **collenchyma** (and parenchyma): living cells, thickened only at the corners with **pectin**, which is flexible like rubber, so the stalk is soft and bends.

2. Why is a thick cuticle useful for a desert plant but a problem for a plant living underwater?

Desert: water is scarce. A thick waxy cuticle cuts down water loss by transpiration, so the plant survives.
Underwater: water is everywhere and there is no risk of drying. The plant takes in water, minerals and dissolved gases directly through its surface, and a thick cuticle would block this. That is why water plants have a thin or no cuticle.

3. How do the "dead" xylem cells work with the living leaf cells to keep water moving up against gravity?

Dead xylem vessels and tracheids are **empty, hollow tubes** with strong lignified walls, joined end to end into a continuous pipe from root to leaf. Living leaf cells lose water by **transpiration** through stomata. As water evaporates, it creates a **transpiration pull** that drags the whole water column up the xylem, since water molecules stick together. So the living cells provide the pull and the dead cells provide a clear, non-collapsing pipe. (Root pressure from below also helps a little.)

4. What would happen if there were no stomata in the epidermis of stems or leaves?

No **gas exchange**: carbon dioxide could not enter for photosynthesis, so the plant could not make food; oxygen could not be exchanged properly. No **transpiration**: no transpiration pull, so water and minerals would not rise well from the roots, and leaves would overheat without the cooling effect. The plant would weaken and die.

5. **In the dance poses of Fig. 3.17, which joints are used and what movement does each allow?**

- **Shoulders and hips** (ball and socket): arms raised overhead or swept in circles, legs lifted sideways or turned out.
- **Elbows, knees and fingers** (hinge): bending of arms in mudras, the bent-knee stance (as in the half-sitting position of Bharatanatyam).
- **Neck** (pivot): side-to-side head movements.
- **Wrists and ankles**: turning the hands in mudras, standing on toes or heels (gliding/hinge-like movement).
- **Skull** (fixed): no movement, protects the brain while dancing.

Revise, Reflect, Refine

1. **Meristematic tissues divide repeatedly. Which property allows this? (i) thick walls (ii) large vacuoles (iii) thin walls, dense cytoplasm, large prominent nucleus (iv) differentiated cells**

(iii) Thin walls, dense cytoplasm and a large prominent nucleus. Thick walls, large vacuoles and differentiation all belong to permanent cells that no longer divide.

2. **If a plant cannot transport food from leaves to roots, which tissue is malfunctioning? (i) Xylem (ii) Phloem (iii) Epidermis (iv) Sclerenchyma**

(ii) Phloem. Its sieve tubes carry food from the leaves to other parts.

3. **Why is the epithelium lining internal organs usually only one or a few cells thick? (i) store food (ii) maximum strength (iii) quick exchange of materials (iv) reduce friction**

(iii) To allow quick exchange of materials across them. A thin layer means a short distance for diffusion (for example, oxygen into the blood in the lungs).

– **4. Compare the straight-leg jump (knees and ankles stiff) with a normal jump (knees and ankles bent). How did ankle, knee and hip positions differ?**



Fig. 3.21 Bent-knee and straight positions

Straight-leg jump: ankles, knees and hips stay nearly straight and locked. Only the toes and calf push a little, so the jump is low, and landing is jarring because the joints cannot absorb the shock.

Normal jump: first the ankles bend, knees bend and hips bend (a crouch, as in the left figure); then all three straighten together with force. The muscles get a longer push, so the jump is higher, and on landing the joints bend again to cushion the impact. The ankle, knee and hip work as hinge-type joints together with muscles and tendons.

– **5. Which type of joint is involved when you bend your knees and ankles? (i) Ball and socket (ii) Hinge (iii) Pivot**

(ii) Hinge. Knees (and ankles, in up-down movement) bend in one plane, like a door hinge.

6. Assertion-Reason. Options: (i) both true, R explains A; (ii) both true, R does not explain A; (iii) A true, R false; (iv) A false, R true.

A. A: Epithelium is well suited for gas exchange in the lungs. R: It has many layers of tall cells that slow diffusion. → **(iii)**. The assertion is true, but lung epithelium is a **single layer of thin, flat cells**; many layers would slow exchange.

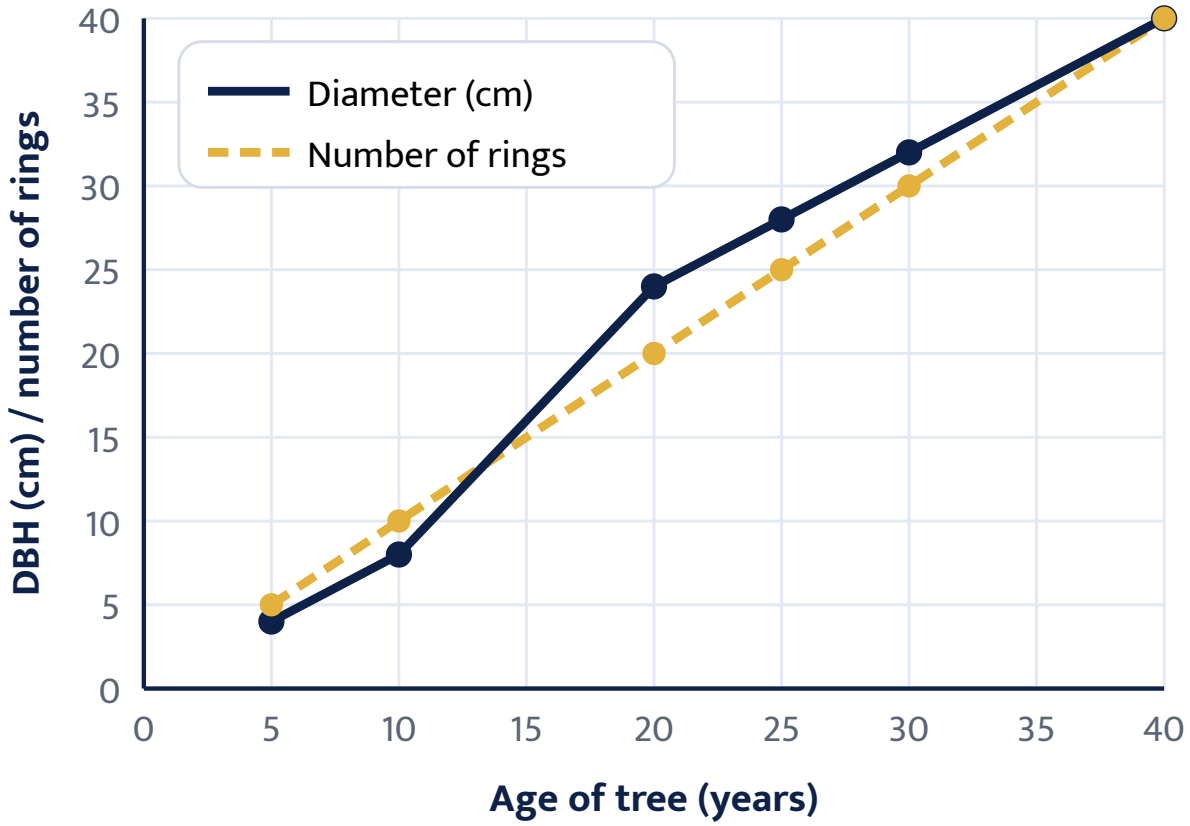
B. A: Cardiac muscle contracts continuously without fatigue. R: Its cells have many mitochondria and a rich blood supply. → **(i)**. Plenty of mitochondria and blood give a constant supply of energy and oxygen, which explains why it does not tire.

C. A: Tendons connect bone to bone and allow joint movement. R: Tendons are tough connective tissue that transmits force from muscle to bone. → **(iv)**. Tendons connect **muscle to bone**; ligaments connect bone to bone. The reason is correct.

D. A: In a hinge joint, movement is mainly in one plane. R: The bone ends are shaped to slide in all directions. → **(iii)**. The bone ends of a hinge joint are shaped to allow movement only in **one plane**; sliding in all directions would not be a hinge.

7. Plot age of a teak tree (x-axis) against diameter (DBH) and number of annual rings (y-axis) from Table 3.7, and interpret.

Age (years)	5	10	20	25	30	40
DBH (cm)	4	8	24	28	32	40
Annual rings	5	10	20	25	30	40



(i) The diameter keeps increasing with age, so the stem grows thicker every year. Growth is not equal every year: it is about 0.8 cm per year from 5 to 10 years, **fastest (1.6 cm per year) between 10 and 20 years**, and about 0.8 cm per year after 20 years. So the tree grew fastest in its young-adult years (or those years had better rain and climate).

(ii) The number of rings is always **equal to the age** (one ring per year), so the rings line is a straight line. Diameter rises as the number of rings rises, but not in exact proportion, because some rings are **wide** (good years, 10–20) and some **narrow**. Diameter = sum of the widths of all rings on both sides.

(iii) The **lateral meristem** (vascular cambium), present as a **ring in the stem** between the xylem and the phloem, is responsible for the increase in girth.

8. **An elephant severely debarked a tree to eat the bark. (i) Which functions are hampered? (ii) Which tissue is affected by further damage? (iii) Which function is hampered if tissues beneath the bark are damaged? (iv) What assumptions are you making?**



Fig. 3.22 An elephant debarking a tree

(i) Protection: the bark (cork) keeps out water loss, germs, insects, fire and injury. Without it the trunk dries and gets infected. Since the phloem lies just inside the bark, part of it is often torn off too, so **food transport** to the roots is also affected.

(ii) The **phloem** first, then the **lateral meristem** (cambium) and the outer **xylem**.

(iii) If phloem is destroyed, **food cannot reach the roots**; the roots starve and slowly die, and then the whole tree dies. If the cambium is destroyed, **growth in girth** stops at that spot and the wound cannot heal. Deeper damage to xylem cuts the **upward flow of water**.

(iv) Assumptions: the bark was removed from only part of the trunk (not a complete ring), and only the outer layers were removed. If bark and phloem were removed **all around** the trunk (a full ring, called girdling), the roots would starve and the tree would surely die. If only a small patch was removed, the tree could survive and form new cork from the cork cambium to heal the wound.

9. **A young mango sapling bends in monsoon winds without breaking. Which tissue gives this flexibility? What if it were replaced by sclerenchyma?**

Collenchyma: living cells thickened only at the corners with pectin, which gives strength with flexibility. If it were replaced by **sclerenchyma** (dead, thick lignified walls), the young stem would become rigid and stiff. In strong winds it could not bend, so it would **snap**. Also, being dead and rigid, it could not stretch as the young stem grows.

10. Sohan planted sugarcane cuttings of type A and type B. Only type B sprouted. Explain.

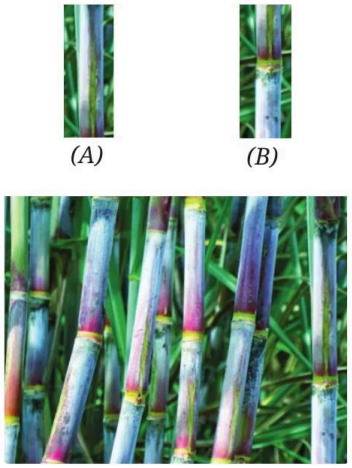


Fig. 3.23 Type A and type B cuttings

(i) Type B cuttings had a **node** with a bud and meristematic (intercalary) tissue, whose cells can divide to form a new shoot and roots. Type A was only a piece of **internode**, made of permanent tissue that cannot divide, so it could not sprout.

(ii) Type B contained a **node** (bud); type A did not.

(iii) Whether the cuttings **sprouted** (new shoots and roots appeared), and how many sprouted, or the length of the shoots after a few weeks.

(iv) Same sugarcane variety and age, same length and thickness of cutting, same soil, water, sunlight and temperature, same planting depth and same number of days.

11. Rohan: "A tissue is a group of similar cells performing similar functions." Rajiv: "True for simple tissues, but different for complex tissues." Explain.

Rajiv is right. In **simple** tissues (like parenchyma) all cells are of one type, so Rohan's definition fits. But **complex** tissues like xylem have **different types of cells** (tracheids, vessels, xylem parenchyma, fibres), some living and some dead, different in shape, all working together for **one common function** (conduction). So a better definition is: a tissue is a group of cells, **similar or different**, that work together to perform a specific function.

12. Coconut husk fibres make tough mats. Which tissue gives this strength? Why couldn't living parenchyma do it?

Sclerenchyma. Its cells are long, dead, tightly packed and have very **thick lignified walls**, so they are hard, strong and do not rot easily. Parenchyma has **thin walls**, loosely packed cells with air spaces and living contents; it is soft, would squash and tear, and its living contents would decay once dried. So it cannot give the toughness needed for mats.

13. Vibha says meristematic cells are found only at root and shoot tips. Is she right? What can Neha ask her?

The statement is **incorrect**. Only **apical** meristems are at the tips. **Lateral** meristem is in a ring in the stem (for girth) and **intercalary** meristem is at the nodes / base of internodes (for regrowth). Neha could ask: "If meristems are only at the tips, then how does a tree trunk become thicker every year, and how does grass grow back after its tips are mowed?"

14. A plant cell and an animal cell are the same size. (i) Which has a larger vacuole? (ii) What are you assuming?

(i) The **plant cell**. A mature plant cell has a large central vacuole (it can fill most of the cell) that stores water and sap and keeps the cell firm (turgid) to support the plant. Animal cells have only small, temporary vacuoles, if any.
(ii) Assumptions: both are **mature, ordinary** cells. If the plant cell were a **meristematic** cell, it would have no vacuole. Some special animal cells (like fat cells) store material in large vesicles.

15. A textbook says "Each plant tissue performs only one specific function." What questions would you ask to check this, and which examples would you use?

Questions to ask: *Does parenchyma only store food? Does the epidermis only protect? Does xylem only conduct water?*
Examples show the statement is **not correct**:

- **Parenchyma**: stores food, photosynthesises in green parts, and gives buoyancy in aquatic plants.
- **Epidermis**: protects, but also has stomata for gas exchange and transpiration, and root hairs for absorption.
- **Xylem**: conducts water and minerals and also gives strength.
- **Phloem parenchyma**: stores food, resin, tannins and latex.

So a tissue has a **main** function but may perform several.

Diagrams from the NCERT Exploration textbook (Grade 9).